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"Science Chariots to Victory."

Editor's Desk

Greetings to all readers of this Newsletter.

A very Happy New Year 2021 to all our readers of this newsletter. We are here with our next issue, which again brings forth interesting articles and information. We now have onboard Dr Vinay Kamble, Former Director of Vigyan Prasara, Govt. of India, as Advisor.

At the outset we wish to make an appeal to all GSA members (Fellows, Associated Fellows and Members) to kindly give your latest contact details and affiliations (at <https://forms.gle/rfy8VH4UnPjxYs5u8>). This is required to be in touch with you and take your guidance and articles for this newsletter.

We are also introducing a column in the newsletter from next issue on Series of Interviews by Eminent Scientists of Gujarat stating their achievement and their journey in the world of science. The column will also highlight the Eminent Women Scientists and Achievers of Gujarat. For this activity, our team will personally invite the achievers.

In our News column of this issue we share with you exciting news. Prof Pankaj Joshi, President of GSA, has been elected as a Fellow of the prestigious 'The World Academy of Sciences (TWAS)'. We are privileged to give details about this. Climate change has resulted in Ocean acidification. This may make some species glow brighter says a study. A study highlights how Catastrophic Events on Earth are imprinted on the Moon. We have a review article on Remote Sensing Applications in Vegetation Studies by Dr Alpna Shukla and an article on Soil Erosion: A Serious Environmental Issue by Dr Benidhar Deshmukh and Rohit Kumar. In Youth Corner, we have two interesting articles by students, Is Nalsarovar Dying? And One Planet: Conversations that matter. Few important Publications also find a mention here.

We again make an appeal to all members to submit articles and announcements. These can be submitted at the newsletter email address gsa.nl.2020@gmail.com. We would also be happy to publish an abstract on new research work and thesis submitted by students in Gujarat. You are requested to give information about the same at gsa.nl.2020@gmail.com.

Enjoy Reading

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News

Ocean acidification may make some species glow brighter



As pH drops, certain bioluminescent species might get extra glow, while others dim the lights

These glowing specks are sea fireflies (*Vargula hilgendorffii*) on a beach in Japan. A new analysis suggests that they might glow a bit brighter as the ocean becomes more acidic.

A more acidic ocean could give some species a glow-up. As the pH of the ocean decreases as a result of climate change, some bioluminescent organisms might get brighter, while others see their lights dim, scientists report January 2 at the virtual annual meeting of the Society for Integrative and Comparative Biology.

Bioluminescence is de rigueur in parts of the ocean (SN: 5/19/20). The ability to light the dark has evolved more than 90 times in different species. As a result, the chemical structures that create bioluminescence vary wildly — from single chains of atoms to massive ringed complexes. With such variability, changes in pH could have unpredictable effects on creatures'

ability to glow (SN: 7/6/10). If fossil fuel emissions continue as they are, average ocean pH is expected to drop from 8.1 to 7.7 by 2100. To find out how bioluminescence might be affected by that decrease, sensory biologist Tom Iwanicki and colleagues at the University of Hawaii at Manoa gathered 49 studies on bioluminescence across nine different phyla. The team then analyzed data from those studies to see how the brightness of the creatures' bioluminescent compounds varied at pH levels from 8.1 to 7.7.

As pH drops, the bioluminescent chemicals in some species, such as the sea pansy (*Renilla reniformis*), increase light production twofold, the data showed. Other compounds, such as those in the sea firefly (*Vargula hilgendorffii*), have modest increases of only about 20 percent. And some species, like the firefly squid (*Watasenia scintillans*), actually appear to have a 70 percent decrease in light production. For the sea firefly — which uses glowing trails to attract mates — a small increase could give it a sexy advantage. But for the firefly squid — which also uses luminescence for communication — low pH and less light might not be a good thing.

Citations

T. Iwanicki, H. DeTurk and M.L. Porter. Turning up the lights: Ocean acidification may increase light intensity of secretory bioluminescent signaling. The Society for Integrative and Comparative Biology Annual Meeting. Presented January 2, 2021.

E.S. Lau and T.H. Oakley. Multi-level convergence of complex traits and the evolution of bioluminescence. *Biological Reviews*. Published online December 11, 2020. doi: 10.1111/brv.12672.

S. Martini and S.H.D. Haddock. Quantification of bioluminescence from the surface to the deep sea demonstrates its predominance as an ecological trait. *Scientific Reports*. Vol. 7. By Bethany Brookshire

Science News, January 11, 2021, by Bethany Brookshire, writer at Science News for Students. She has a Ph.D. in physiology and pharmacology from Wake Forest University School of Medicine.

Gujarat Black Hole Scientist Elected TWAS Fellow

Noted Astrophysicist and Cosmologist Dr. Pankaj S Joshi (Vice-Chancellor (Provost) and Founding Director, International Center for Cosmology (ICC), Charusat University (Charotar University of Science and Technology), India), has been elected as a Fellow of the prestigious The World Academy of Sciences (TWAS). TWAS based in Italy has been working on how Developing Countries can address challenges like hunger, disease and poverty by strengthening science and engineering. Professor Joshi will be inducted in a ceremony to be held during the next general meeting of TWAS.



He has made fundamental contributions in black hole physics, gravitation and cosmology, which have brought him international acclaim. His extensive analysis of general relativistic gravitational collapse has been widely recognized as providing key significant insights into the final fate of massive collapsing stars in universe, space-time singularities and cosmic censorship. Over past years, his paradigm changing work has shown that massive stars could collapse to black holes, or exploding fireballs called naked singularities. In later case, observable signals from ultra-dense regions near a singularity may reach faraway observers in the universe. Astrophysical and quantum gravity signatures of the same are being explored now. His research, some 200 publications in reputed International journals, books and monographs in topmost series from the OUP (Oxford) and CUP (Cambridge) are now widely cited, making this a frontier research field in black hole physics and cosmology. Member of the Indian National Science Academy, New Delhi and National Academy of Sciences, India, he has received the DAE C. V. Raman lecture award and the Prof. A. C. Banerji Gold Medal and Memorial Lecture Award, among others. (From <https://twas.org/>)

Catastrophic Events on Earth Imprinted on the Moon

Thousands of tiny meteorites hit the surface of the Earth each year. Most of them remain unnoticed; big meteors are rare. However, those that survive the caustic passage through our atmosphere potentially threaten life. Many scientists have correlated significant geology or climate changes and catastrophic events, like mass extinctions, with bolide impacts. The best-known example is the dinosaur killer asteroid that shocked Chicxulub, Mexico, about 65 million years ago terminating the Cretaceous-Paleogene era —giving a chance to the mammals and, later, the humans to dominate.

After a significant meteorite hits our planet, the evidence of the impact spreads all over the place. However, the Earth is alive, it is made out of molten rocks with a moving crust on top, tectonic plates colliding, volcanoes erupting, and lava becoming ocean floor. Weathering and resurfacing prevent the preservation of most craters. All this restlessness erases the evidence of very old impacts. For planetary scientists, the number, size, shape, and composition of the expelled material of meteorite impact craters tell the history of celestial bodies.

Because the Moon has been evolving together with the Earth for billions of years, meteorite bombardments affect both simultaneously. The Earth is a bigger target than the Moon, and it is certainly more attractive for cosmic bodies, gravitationally speaking. Nevertheless, the Moon suffers much less erosion than the Earth. The geology of its surface is practically an account of meteorite fall. In Japan, a research group led by Professor Kentaro Terada, Osaka University, investigates the formation and age of 59 large craters on the Moon using data from the Kaguya satellite. In a recent study published in Nature, they showed that meteoroid showers bombarded the Earth-Moon system approximately 800 million years ago, right before the onset of the first known glacial period — a moment of significant climate and biological changes. Geologists believe that the entire planet turned into an iceball for about 85 million years. They named this period the Cryogenian. The team found that several craters should have formed simultaneously. They remarked that their most important finding, which is the key to the novel discovery, is that eight craters

bigger than twenty kilometers showed identical ages. Professor Kentaro Terada and collaborators estimated that at least ten trillion tons of extraterrestrial rocks, 30 to 60 times more than the Chicxulub impact, hit the Earth at that moment. They suggested that the vast amount of space material they estimated to have entered the Earth 800 million years ago may have triggered some of the significant changes in climate and life, giving way to new species, in the Cryogenian period.

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<https://www.ua-magazine.com>

Remote Sensing Applications in Vegetation Studies: A Review

Abstract:

Mapping vegetation through remotely sensed images involves various considerations, processes and techniques. Increasing availability of remotely sensed images due to the rapid advancement of remote sensing technology expands the horizon of our choices of imagery sources. Various sources of imagery are known for their differences in spectral, spatial, radioactive and temporal characteristics and thus are suitable for different purposes of vegetation mapping. Generally, it needs to develop a vegetation classification at first for classifying and mapping vegetation cover from remote sensed images either at a community level or species level. Then, correlations of the vegetation types (communities or species) within this classification system with visible spectral characteristics of remote sensed imagery have to be identified. These spectral classes of the imagery are finally translated into the vegetation types in the image interpretation process, which is also called image processing. This paper presents an overview of how to use remote

sensing imagery to classify and map vegetation cover. This paper focuses on the comparisons of popular remote sensing sensors and their utilization for studying various aspects of vegetation.

Keywords: vegetation mapping, remote sensing sensors, image classification and its techniques

Introduction

The technology of remote sensing offers a practical and economic study of vegetation cover and its changes over the large area. Because of its capability Classifying and mapping vegetation is an important technical task for managing natural resources as vegetation provides a base for all living beings and plays an essential role in affecting global climate change. Current remote measurement capabilities provide an opportunity for fundamentally new approaches to the classification of functional types, which could even lead to a redefinition of the term itself. Satellite imagery has proved to be very effective in both mapping and measuring areas in vegetation monitoring and different plant species, mapping includes height, density, and extent of dead and dying timber.

Several important factors must be considered in designing remote sensing instruments for this purpose. Some of the most important are:

- (a) **Spectral bandwidth** - The bandwidth is normally selected to be wide enough to obtain sufficient energy and narrow enough to detect the spectral features of interest.
- (b) **Spectral selection** - If complete spectrum sampling is not employed; the bands will be placed to be coincident with spectral features of interest.
- (c) **Spectral resolution** - This is determined by the spectral spacing between band centers, not by the spectral bandwidth.
- (d) **Temporal resolution** - The time of an image or the time between two images. The question is whether the instrumentation is to be used for routine, repeated monitoring, or to provide information on episodic events. The answer will be a factor in determining the appropriate orbit of the satellite and the timing of the imagery.
- (e) **Intensity resolution** - The utility of finer and finer intervals of measurement of intensity is ultimately limited by the noise of the sensor itself and by the inherent variability in the scene.

(f) **Accuracy** - The atmosphere often interferes with reception. In addition, there may be a need for absolute calibration of the instrument in addition to normalizing between spectral bands.

(g) **Correlation with other data or maps** - Remotely sensed information will seldom give all the information required. The data must be prepared and transmitted to the user in a form that is useful in comparisons with other data.

(h) **Data delay** - Thoughtful instrument design may allow the production of data in forms that minimize the time needed to process it on the ground. A large part of the instrument designer's and scientists joint task is mutually to determine a suitable world model which can be solved using available data, and the sensor system to obtain that data.

Global vegetation databases have been developed from published maps, atlases, and national databases that attempt to represent existing vegetation (Matthews, 1983). These databases provide global models with a generally realistic estimate of current land cover at coarse spatial resolution. However, these databases suffer from lack of consistency in vegetation classification used, variable measurement techniques, and a variety of spatial sampling resolutions. Consequently, Townshend et al. (1991) argued that the most essential new global vegetation classification must be remote sensing driven, to provide a realistic measure of existing land cover. However, current remote sensing capabilities cannot produce the large number of land cover classes usually defined, particularly because climate classes are usually part of the definition (Loveland et al., 1991). There is no clearly defined set of vegetation characteristics used for these classifications; there is significant disagreement among authors of the existing global extent of different biome classes. A simpler logic based only on clearly observable plant physiognomic characteristics is needed. As our need for ecosystem and global approaches to biodiversity conservation increases, it becomes increasingly important to use remote-sensing tools to characterize landscapes in biologically meaningful ways.

Remote sensing has been used to map a variety of ecosystem patterns and processes correlated with vegetation type in the Arctic, e.g., biomass (Shippert et al. 1995), composition (Shippert et al. 1995), thaw depth (Nelson et al. 1997), and CO₂ flux (Stow et al. 1998). An important question in these studies is how well we can predict ecosystem patterns and processes from remote sensing data. The normalized difference vegetation index (NDVI) is the most widely used index for remote sensing of vegetation in the past two decades. It is equal to $(NIR-R)/(NIR+R)$, where R is the radiance (in reflectance units) of a red channel near 0.66 μ m, and NIR the radiance (in reflectance units) of a near-IR channel around 0.86 μ m. Numerous other vegetation indices with varying complexity have been developed using the same set of near-IR and red channels during the past decade. These indices do not have nearly the same popularity as that of NDVI.

Remote Sensing Sensors

Landsat TM
Landsat ETM
SPOT
MODIS
IKONOS
Quickbird
IRS-Series
Sentinel-1A
Sentinel-2A

Role of Optical Sensors in Vegetation Monitoring

Optical Remote Sensing utilizes the Visible, NIR and SWIR sensors to generate images of the surface of the earth on the basis of detection of reflection of solar radiation from the objects on the ground. Each and every material has distinct property of reflection and absorption which is different at different wavelengths. Therefore, each material when viewed in a remotely sensed image has a different spectral signature. Classification of remotely sensed sensors is as follows (Principles of Remote Sensing):

a) **Panchromatic Sensors:** These sensors measure the brightness of the target and the colour of the target is lost in the image. Examples are IKONOS-PAN, SPOT HRV-PAN, CARTOSAT, etc.

b) Multispectral Imaging Sensors: These sensors generate images which contain both brightness and colour of the targets and have few spectral bands. Examples include LANDSAT TM, IKONOS MS, RESOURCESAT-2 LISS-4 etc.

c) Superspectral Imaging Sensors : These sensors have more channels than multispectral sensors mostly greater than 10 channels. Examples are MODIS, MERIS etc.

d) Hyperspectral Imaging Sensors : These sensors have more than 100 channels and have multifaceted applications. AVIRIS and Hyperion are the best examples.

The basic parameters that should be taken into consideration for vegetation monitoring using optical sensors are as follows:

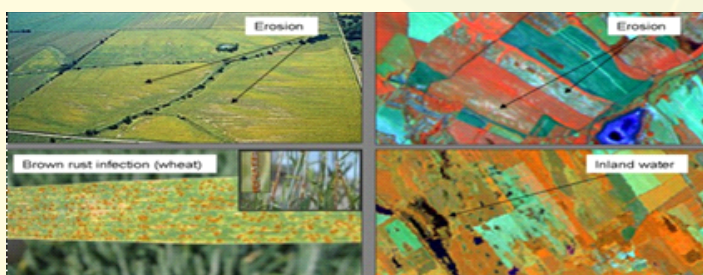
a) Spectral sampling: This requires consideration of various bands for study of vegetation. The bands of 0.35 to 0.43 μm are suitable for atmospheric scattering and haze. The different bands from 0.43-0.8 μm are suitable for various characteristics of chlorophyll.

b) Spatial Sampling: Noise in the scene is a measure of texture. Texture is ordinarily described by the characteristics such as smooth, fine, rough etc. This parameter takes into consideration the spatial distribution of the pixels.

c) Temporal Sampling: This factor deal with the availability of data at various time intervals depending on the feature of vegetation under study accordingly the type of data should be used. Increased repetivity of the sensors within a short period can be used to study the change in signature.

d) Accuracy of Data: The absolute calibration of the instrument must be done to ensure that the reflectance and radiance values given by the pixels of image are proper.

e) Correlation with other data: Comparison with other data is necessary to check the variation patterns.



Role of Microwave Sensors in Vegetation Monitoring

Microwave remote sensing has played a very important role in monitoring the earth and its numerous applications and proved to be a great tool in developing several techniques responsible for maintaining an ecological balance between nature and humans. Microwave remote sensing is very useful for forestry applications due to their capability to penetrate the forest canopy and contribute to the monitoring of the forest and help in understanding the ecosystem processes.

Microwave radiation is capable of penetrating significant distances into vegetation canopy and it interacts strongly with structures (such as leaves, stems, etc) on a large scale comparable with the wavelength of the radiation.

A combination of wavelength and polarization characteristics decides the penetration depth of the radar waves into the canopy.

Depending on these two factors sensing of different plants of different sizes, shapes and water content is possible. This powerful ability of radar to reach various levels of canopy provides a great motivation for forest research. (Sun et. al. 1998). In addition, the sensor parameters such as incidence angle and wavelength of the radar are also responsible for particular signatures on the radar images. It is drawn clearly that there is a relationship between wavelength such as (C, L, P band) radar backscatter and biomass of forest and the growth in stock volume. (Tansley et. al. 2004). Therefore, these particular bands are suitable for the biophysical parameter retrieval.

Carbon Accounting and Biomass estimation are the most important applications of radar remote sensing. Also the main applications of Synthetic Aperture Radar related to vegetation monitoring are:

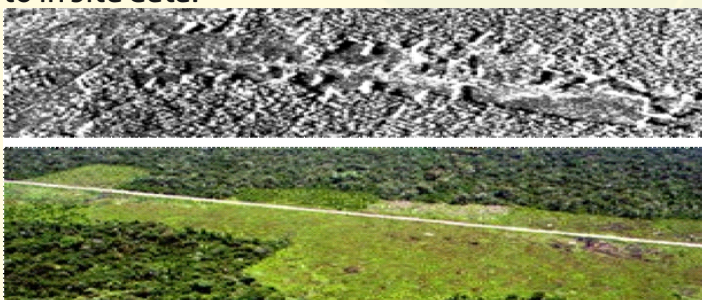
- Studying the characteristics of soil and vegetation cover*
- Inundation below closed canopy*
- Fresh woody biomass of forested areas*
- Freeze/Thaw conditions of soil and vegetation*
- Soil moisture and surface roughness in low vegetation areas*
- Terrain surface and vegetation canopies (Heri et. al.)*

Vegetation including trees is quite rough for longer wavelengths. This increase in roughness leads to increase in brightness in the radar image. Tree structures affect the radar backscatter as backscattering and penetration varies within forest canopy. Shorter wavelengths cannot penetrate hence they are just limited to the tree tops whereas at longer wavelengths the penetration is high. With the use of Digital Terrain models, tree height estimation can be done. Scattering properties are determined by the size, structure and shape of the tree within the canopy (Floyd et.al 1998). The parameters that are important in forest inventory are tree density, stand age and timber volume. Structures of trees affect the backscattering coefficient (Touzi et al.,2004).

Discrimination between different types of land cover classes is done with the help of backscatter intensity and texture. Image fusion techniques and Polarimetric decomposition techniques can be utilised for this purpose. Changes in tone and texture are related to crown closure or foliage density. Radar backscatter is dependent on crown closure than height (Floyd et. al. 1998). Radar backscatter depends on other factors such as electrical properties of target i.e. dielectric content which represents the water content in the target. Estimation of total biomass can be easily done with the help of cross polarized images i.e. HV, VH as tendency of microwaves to get depolarized is very high when they interact with the forest canopy.

Underlying surface conditions can also be determined by the unique ability of SAR to discriminate between herbaceous biomass from woody biomass.

Radar waves are very much sensitive to parameters of forest such as girth at breast height (GBH) and tree mean height. The Interferometric SAR (InSAR) technique has a great potential in providing stem volume estimates in comparison to in situ data.



Conclusion

Various aspects of vegetation monitoring with respect to optical and microwave remote sensing were discussed in this paper. Depending on the type of feature of vegetation under study a particular type of sensor should be used and in many cases, the combination of microwave and optical sensor images can be utilised to develop fusion techniques in order to segregate the features.

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Article Written By : Dr. Alpna Shukla, Head
(Botany Department, M.G. Science Institute)

Soil Erosion: A Serious Environmental Issue

Soil erosion and its effects

Soil is one of the most important natural resources essential for sustaining our life on earth as it provides us food. As reported by Food and Agriculture Organization of the United Nations (FAO) in its 2015 report, more than 95% contribution of world food production is from soil. However, in the 21st century, soil erosion is considered as a key or gravest threat to soils (Amundson et al. 2015; Montanarella et al. 2016) and is one of the most important processes occurring on the surface of the earth (NRSC Report, 2014). In the process of soil erosion, soil particles are detached, which are transported and then deposited elsewhere through geomorphic agents such as wind and water. Soil degradation due to soil erosion is a threat to sustainable development and growth as it results in low crop production because the eroded fertile top layer is rich in nutrients which are required by plants (Poesen, 2018). It also results in suspension of sediments in water leading to accumulation of

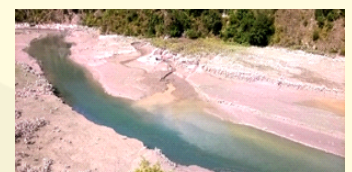
sediments (Fig.1) and clogging of waterways and infrastructure such as drainage networks, embankments and dams affecting their efficiency. The loosened soil particles may increase air pollution and also aid in thereby leading to change in the ecosystem and loss of biodiversity. It is believed that due to expansion of agricultural lands the rate of soil erosion is increasing and the adverse effect may be witnessed more in the least developed economies (Borrelli et al. 2017). There are now several studies that indicate that soil erosion is a major cause of land degradation. Therefore, it has now become a serious environmental issue requiring attention of researchers and policy makers.

Causes of soil erosion

Soil erosion is a natural continuous process but it is exacerbated by anthropogenic activities such as deforestation and also agricultural/farming practices in particular, for which about 40% of land is being utilised (Foley, 2017). When vegetation is removed it loosens the soil and makes it dry and vulnerable to erode. Besides the deforestation and agriculture, there are many other causes of soil erosion such as rainfall, overgrazing, mining, movement of animals, etc. (Fig.1). Generally, rainfall and flowing/running water are the most common agents driving soil erosion. Most of us have come across the evidences of soil erosion in road sides after rainfall. Similarly, you might have also noticed that when we water the flower pots or in vegetable garden there is little amount of soil that is washed away. Another form of moving water but in solid state are glaciers that also move a large quantify of soils and rocks. Even construction and some of the recreational activities are also responsible for soil erosion (Fig.1d-f). Soil erosion takes place not only in wet areas but also in areas which are dry and having little or no vegetation, wherein it primarily takes place due to the agent i.e. wind that blows away lighter soil particles.



(a)



(b)

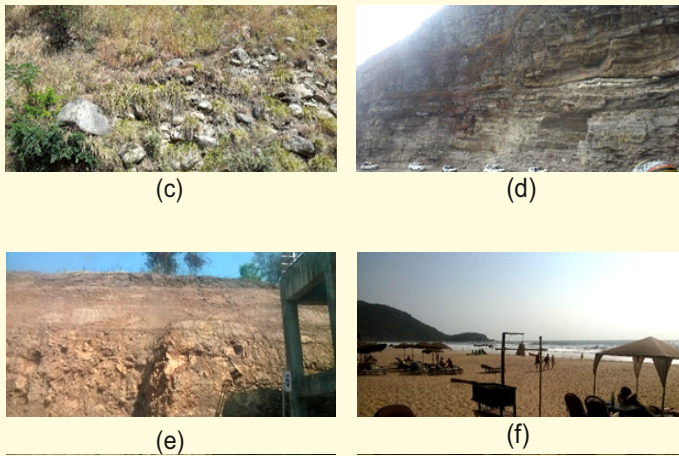


Figure 1: Some examples showing soil erosion caused by rainfall and running water (a) at road side, (b) sediments in suspension in river water, and (c) sloping land; due to anthropogenic activities such as construction of (d) road in hilly areas, (e) rail route cutting, and (f) recreational activities at a beach.

Types of soil erosion

Generally, five types of soil erosion as given in the table 1 are recognised:

Table 1: Types of soil erosion

Type	Agent	Effect/Process
Splash erosion	Rain drops	When raindrops fall on the earth surface, it causes splashing of soil particles in different directions and thus forming small depressions
Sheet erosion	Runoff flow	Moving water in a large sloping area results in uniform removal of top soil however it is not that easy to discern
Rill erosion	Rapid concentrated runoff flow	Moving water results in non-uniform removal of top soil causing formation of small channels, which are known as rills
Gully erosion	Rapid runoff flow	Rapid runoff flow causes expansion and deepening of rills thereby forming features, which are known as gullies
River bank /channel erosion	Rapid runoff flow	It occurs due to high velocity runoff flows and clearing of vegetation from the river bank causing instability of the banks and their wearing away

Protection from soil erosion and the role of soil erosion modelling

We know that to sustain our life we need agriculture produce which in turn is possible only when there is availability of fertile soils. Soil erosion is washing away the fertile soils causing lower agriculture productivity. According to the reports of Food and Agriculture Organisation (FAO) published in 2015 and 2017, most of the world's soil resources are in either fair or poor or in very poor condition and soil erosion due to water and wind are the major threats globally to the soils and their ecosystem services. Further, it has also been recognised that the current rate of soil erosion is higher than the rate of new soil formation or replenishment so, the fertile soils may be lost after some period of time and also it

will reduce agriculture productivity (Amundson et al. 2015). Therefore, it is essential to reduce soil erosion by adopting some conservation practices. However, prior to adopting any soil conservation strategy it is required to identify the areas undergoing erosion and the rate of erosion and then only a suitable mitigation measure to reduce erosion can be effectively implemented.

It is now possible to identify the areas undergoing soil erosion, estimate the rate of erosion and also to test the possible effects of chosen mitigation measures through soil erosion modelling, which may provide outputs at different spatial and temporal scales. This has become possible due to efforts of number of scientists to develop tools and techniques for studying soil erosion caused particularly by water. Oldest efforts can be traced back to around mid 19th century. Advent and availability of spatial data acquisition and processing technologies such as remote sensing and Geographic Information System (GIS) has been advantages in soil erosion studies.

There are many models available for modelling and predicting soil erosion, which can be categorised into empirical, conceptual and physically based or process oriented. There are many models but the most popular and commonly used is the USLE (Universal Soil Loss Equation) possibly due to

- i) its perceived simplicity,
- ii) the ease in application, and
- iii) applicability in wide variety of regions.

The USLE model estimates the annual average soil erosion considering several parameters such as energy and amount of rainfall, soil erosion potential, slope length and steepness, land cover and soil conservation or management (Wischmeier and Smith, 1965). The USLE was originally developed as an empirical model for the agriculture field plots of USA to support management decision. This model has been modified by different researchers and process oriented components have also been incorporated so as to enhance its applicability in different regions of the world at different spatial scales. However, one of the limitations of this model is it does not account for the gully and stream bank erosion. Despite the limitation, the USLE type model has been found useful in conservation planning as it provides the planners

a tool for the assessment of soil erosion at different spatial scales because only then it can be taken into consideration in environmental and agriculture policy programmes and also in integrated land surface models (Alewell et al. 2019).

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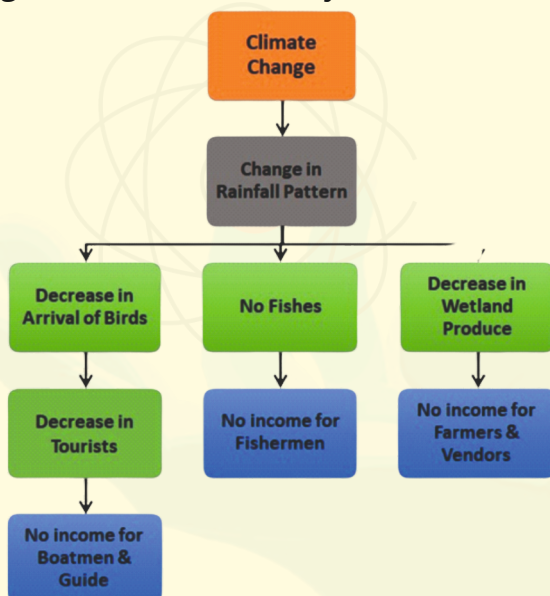
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IS NAL SAROVAR DYING?

The largest bird sanctuary in India, popularly known as Nal Sarovar Bird Sanctuary (NSBS) lies in the semi-arid area, between Ahmedabad and Surendranagar, in the state of Gujarat. It is referred to as a "Wetland". In simple terms, Wetland is an area that is filled with water most of the year or has an on and off water supply. This wetland sanctuary, known for its magnificent landscapes and captivating flora and fauna, attracts a large number of visitors year-round. Unfortunately, considering its present scenario, Nal Sarovar sanctuary seems to be degrading and thus calls for more attention.

Nal Sarovar is a typical natural largest shallow water lake in India, covering an area of 120 sq. km. In 2012, it was declared as one of the 26 Ramsar Convention sites (a global treaty signed between nations to prevent wetland degradation) of India. It is home to many aquatic plants and animals, supplying an exclusive environment for their survival. Also, there are about 12 villages located in the vicinity of the wetland, and certainly, it is the major source of livelihood for the locals. NSBS shelters nearly 216 species of birds, including about 150-wetland dependent species. A lot of these are migratory birds, who travel during winters in October (stays until March), in order to find more abundant food and a better climate. The suitable water level for the sustenance of migratory birds in the NSBS is supposed to be 3-3.5 feet. But something strange was witnessed in the year 2018 when these birds didn't arrive at the lake as it dried up, due to very less rainfall. Furthermore, even the excess water

from the Narmada was not released into the lake by the government, as it was diverted for irrigation of rabi crops. On the contrary, in the following year, the lake was flooded with water due to heavy rainfall and release of surplus water from the Narmada canal which led to water level rising excessively up to 9 feet, and as a result, the migratory birds did not turn up again in October 2019, their usual arrival time. This scenario clearly hints at how changes in the climate and unsophisticated water management are leading to an unstable ecosystem. The local inhabitants, being an inevitable part of this ecosystem, their occupation is also majorly impacted. Their source of income could be from taking tourists for boating, guiding them through the tour & helping them identify a variety of birds. A large population of the community is into fishing, for selling, and to feed their families. Thus, drought conditions affect their living. Many villagers are dependent on the wetland produce for their income. Gondro, Theg, Lapa are some of the major wetland produce which are used to feed livestock, to sell in the market, and for their own consumption too. Due to changes in the rainfall patterns, these produce become inaccessible. It seems that in this interwoven relationship between the locals, Nal Sarovar lake, and climate change, the latter one clearly dominates.



While farming, the farmers use pesticides which are reported to be harmful to themselves and the birds. Arzoo Malik, in The Rufford Foundation report (2018) mentioned that after interviewing 113 farmers it was found that they do experience eye irritation, skin burning sensation, and nausea which occurs while spraying the pesticides in

fields. These pesticides can affect the behavior of birds, change in feeding & mating behavior, decreased egg production, malformed beaks, defects in feather growth, decreased immunity, and likewise health issues. This calls for an urgent need for awareness among the farmers as they are oblivious to the sustainable alternatives like using bio-pesticides which are safe and possess no risk to the environment.

Learning about how important this wetland ecosystem is for the survival of birds and supporting the livelihood of locals, it is an absolute upsetting scenario that NSBS is gradually losing its ability to support them and is degrading every passing year. If not taken care of, the change in climate can drive ecological, societal, and economic collapse in the area of Nal Sarovar in coming future

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One Planet : Conversations That Matter

"The Earth is what we all have in common"

-Wendell Berry

The Environment Club of Ahmedabad University is a community of individuals who are concerned about environmental problems and wish to take a stand on the issue or do something about it. We wish to bring young people on board with our activities to educate them, raise awareness, and bust common myths.

We even believe that taking a small step towards protection of the environment matters a lot, because every little step counts and makes a huge difference. The club is dedicated in organising events in form of webinars, panel discussions, fun games, awareness programs, plantation drives, days celebration and many more; all with the goal of educating the youth, bursting out the common myths and spreading awareness related to the environment. To have an open conversation on issues related to the environment and nature, we came up with the series '**One Planet: Conversations that Matter**' with experts who are actively working in the field of environment and sustainability. To reach a wider public under the pandemic situations, the interviews were pre-recorded and then released

One Planet: Conversations That Matter

- Dr.Sara Ahmed**

 Talked about the current Water Ethics in the context of the heritage Living Water Museum
- Dr.Murari Jha Kumar**

 Discussed about the history of Environment during the Mughal Empire and River Ganga
- Bhargav Patel**

 Talked about his Journey of Being a Young Reporter for the Environment
- Dr.Supratim Das Gupta**

 Spoke about using more solar and wind power instead of coal for generating electricity.
- Mr.Ripu Daman Bevli**

 Told his Life Story of becoming a Plogman to make India litter free
- Mr. Anand Raaj**

 Spoke about his Journey of Starting a Zero-Waste juicebar
- Ms. Gauri Maulekhi**

 Expressed her idea about Animal Welfare and their Rights
- Mr. Ricky Kej**

 Articulated on how music can be useful in spreading environmental awareness
- Dr.Rajgopalan Vasudevan**

 Narrated on how he conquered to make plastic roads and do waste management
- Mr.Kapildeep Agarwal**

 Briefed the Importance of Forest and his role in the protest of Save Aarav Forest

on the club's Instagram handle and are also available on the club's website (<https://sites.google.com/view/ahduni-environmentclub>). The main focus of the talk series was on identifying and appreciating the efforts of the people who came up with an innovative idea for solving the diverse environmental issues. Throughout the series, we came in contact with 10 unique individuals, each from a different walk of life, ranging from a college student to entrepreneurs, and artists and distinguished scientists and professors. Among all, the speakers, the speakers who got into the maximum limelight were Mr. Anand Raaj, Mr. Ricky Kej, Mr. Ripu Daman Bevli, Ms. Gauri Maulekhi, Dr. Rajgopalan Vasudevan and Mr. Kapildeep Agarwal.

Mr. Anand Raaj, popularly known as a 'waste engineer', who worked as a RJ and turned into an entrepreneur talks about the Journey of starting India's first Zero-waste Juice Bar. He took over his father's juice bar business and is setting an example by eliminating the use of single-use plastic glasses and promoting zero-waste practices.

Mr. Ricky Kej, known for his Environmental Music, is an internationally renowned Indian music composer, environmentalist and professor. His passion for music and dedication to protect the environment made him compose music on creating environmental awareness. For his extraordinary efforts, he has been awarded Grammy Awards and titled as #1 artist on a US billboard

Mr. Ripu Daman Bevli is called as the Plogman of India,. Mr. Bevli is an engineer by education and an Environmentalist, Social Activist and an Entrepreneur by passion. After initially working with the IT companies, he started the "Ploggers of India" campaign in 2017. His campaign, "Run to make India Litter-free" was successful in cleaning 50 cities across India.

Ms. Gauri Maulekhi, Trustee of India's largest Animal Welfare Organization - People for Animals, discussed about Animal Welfare and their rights. She has worked with various ministries of the Government of India and to formulate and implement policies, such as humane transportation of animals, humane slaughter practices, effective population management and the protection of wildlife. She believes that protecting animals from malpractices like slaughtering and conserving their rights provided her a purpose for life.

Interview of **Dr. Rajagopalan Vasudevan**, a professor by profession and a scientist by passion was about A Step towards Waste Management. He came up with an innovative idea of reusing plastic waste to construct better, more durable and very cost-effective roads. By shredding plastic waste, mixing it with bitumen and then using the polymerized mix in road construction will help in making roads much faster and also will save the environment from toxic plastic waste. Due to his thoughtful efforts in this area, he has been entitled as 'Plastic Man Of India' and has been awarded Padma Shri in 2018.

Mr. Kapildeep Agarwal's interview was about The importance of Environment and his role in the protest of Save Aaray Forest. He is the current head of environmental issues and national cabinet member of Youth Swaraj. He has been actively participating in environment movements since his graduation.

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- 2) Miracle Rindani** (Coordinating Committee Member at The Environment Club, Ahmedabad University.)



Important list of Publications

Gaebler, C., Wang, Z., Lorenzi, J.C.C. et al. Evolution of antibody immunity to SARS-CoV-2. Nature (2021). <https://doi.org/10.1038/s41586-021-03207-w>

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